

StockLens

AI-Assisted Stock Analysis and Ranking Platform

Team Name: StockLens

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Problem and Target Users

Beginner investors often struggle to understand how stock prices, financial news, and market events relate to investment decisions. Relevant information is scattered across different platforms, and financial indicators can be difficult to interpret without prior experience. StockLens is designed for beginner retail investors and students who want an accessible way to compare stocks and understand the factors behind their rankings.

Proposed Product

StockLens is a web-based decision-support platform that brings market data, financial news, and stock analysis into one interactive dashboard. The system will retrieve data through external APIs, organize it by company and date, and generate stock scores using price momentum, news sentiment, event relevance, and risk indicators. An AI component will analyze news and provide plain-language explanations grounded in the retrieved information and calculated scores.

Core Features

- **Integrated Data:** Collect and organize stock prices, trading volume, company information, and relevant financial news.
- **Stock Scoring and Ranking:** Combine normalized indicators into an overall score and rank stocks within the selected universe. Factor weights will be documented and evaluated during development.
- **Interactive Dashboard:** Display a ranking table, stock comparisons, historical price charts, and individual stock details.
- **AI Explanations:** Explain why a stock ranks higher or lower, identify the main contributing factors, and connect explanations to supporting news and indicators.

Typical Use Case

A student with limited investing experience opens StockLens to compare several companies. The student reviews their rankings, selects a stock, and sees its price trend, relevant news, and an explanation of its score. The student can also review historical price trends to understand how the stock has changed over time.

Project Scope and Expected Outcome

The initial prototype will cover a predefined set of up to 100 U.S. stocks, using daily market data and available financial news. Historical stock data and relevant financial information will be used throughout the design and development of our product, drawing on earlier periods where data are available. The system will integrate data collection, processing, scoring, AI explanations, and visualization into a working dashboard. Evaluation will consider data completeness, consistency of score calculations, and whether users can understand the displayed results. StockLens will support learning and stock comparison; automated trading and personalized portfolio management are outside the initial scope. Historical backtesting may be considered as a future extension.